Homework #5

Homework 2

Problem 1

Virtual memory uses a page table to track the mapping of virtual addresses to physical addresses. This exercise shows how this table must be updated as addresses are accessed. The following table is a stream of virtual addresses as seen on a system. Assume 4 KiB pages, a 4-entry fully associative TLB, and true LRU replacement. If pages must be brought in from disk, increment the next largest page number.

4095, 31272, 15789, 15000, 7193, 4096, 8912

TLB

Valid	Tag	Physical Page Number
1	11	12
1	7	4
1	3	6
0	4	9

Page Table

	Valid	Physical page or in disk
0	1	5
1	0	Disk
2	0	Disk
3	1	6
4	1	9
5	1	11
6	0	Disk
7	1	4
8	0	Disk
9	0	Disk
10	1	3
11	1	12

a) Given the address stream in the table, and the initial TLB and page table states shown above, show the final state of the system. Also list for each reference if it is a hit in the TLB, a hit in the page table, or a page fault.

b) Repeat Exercise a), but this time use 16 KiB pages instead of 4 KiB pages. What would be some of the advantages of having a larger page size? What are some of the disadvantages?

c) Show the final contents of the TLB if it is 2-way set associative. Also show the contents of the TLB if it is direct mapped. Discuss the importance of having a TLB to high performance. How would virtual memory accesses be handled if there were no TLB?

解: (a) 4-KiB = 4096 bit

Address	4095	31272	15789	15000	7193	4096	8912
VPN	0	7	3	3	1	1	2

1. Ref 4095 (VPN 0), TLB miss, Page Table hit
2. Ref 31272 (VPN 7), TLB hit
3. Ref 15789 (VPN 3), TLB hit
4. Ref 15000 (VPN 3), TLB hit
5. Ref 7193 (VPN 1), TLB miss, Page Table miss, Page fault
6. Ref 4096 (VPN 1), TLB hit
7. Ref 8192 (VPN 2), TLB miss, Page Table miss, Page fault

Final TLB:

Valid	Tag	PPN
1	1	13
1	7	4
1	3	6
1	2	14

Final Page Table:

	Valid	PPN
0	1	5
1	1	13
2	1	14
3	1	6
4	1	9
5	1	11
6	0	DISK
7	1	4
8	0	DISK
9	0	DISK
10	1	3
11	1	12

(b) 16-KiB = 16384

Address	4095	31272	15789	15000	7193	4096	8912
VPN	0	1	0	0	0	0	0

1. Ref 4095 (VPN 0), TLB miss, Page Table hit
2. Ref 31272 (VPN 1), TLB miss, Page Table miss, Page fault
3. Ref 15789 (VPN 0), TLB hit (假设此处 Tag 11 LRU)
4. Ref 15000 (VPN 0), TLB hit
5. Ref 7193 (VPN 0), TLB hit
6. Ref 4096 (VPN 01), TLB hit
7. Ref 8192 (VPN 0), TLB hit

Final TLB:

Valid	Tag	PPN
1	1	13
1	7	4
1	3	6
1	0	5

Final Page Table:

	Valid	PPN
0	1	5
1	1	13
2	0	Disk
3	1	6
4	1	9
5	1	11
6	0	Disk
7	1	4
8	0	Disk
9	0	Disk
10	1	3
11	1	12

Advantages: ①减少页表大小 ②增大TLB命中率 ③提高I/O效率

Disadvantages: ①所需内存不是页的整数倍，最后一页浪费空间更多

②加载16-KB页更慢，缺页开销大

(c) 2-way set associative (Index=VPN%2, Tag=VPN/2)

Address	4095	31272	15789	15000	7193	4096	8912
VPN	0	7	3	3	1	1	2
Index	0	1	1	1	1	1	0
Tag	0	3	1	1	0	0	1

Final TLB:

Index	Valid	Tag	PPN
0	1	0	5
0	1	1	14
1	1	0	13
1	1	1	6

Final TLB:

Index	Valid	Tag	PPN
0	1	0	5
1	1	0	13
2	1	0	14
3	1	0	6

Direct mapped (Index=VPN%4 Tag=VPN/4)

Address	4095	31272	15789	15000	7193	4096	8912
VPN	0	7	3	3	1	1	2
Index	0	3	3	3	1	1	2
Tag	0	1	0	0	0	0	0

1. 若没有TLB，CPU每次访问内存都需两次访问，降低性能
2. 在没有TLB时，首先从STPBR中获取页表基地址，然后逐级访问页表读取页表项，有效则读取PPN，无效则触发Page Fault，从磁盘读取最终将PPN与页内偏移量offset结合，得到物理地址并读取

There are several parameters that impact the overall size of the page table. Listed below are several key page table parameters.

Virtual address size	Page size	Page table entry size
32 bits	4 KiB	8 bytes

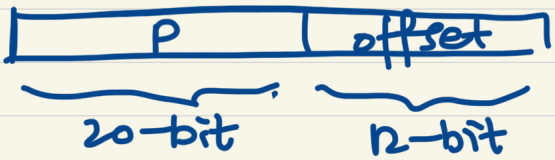
2^{12}

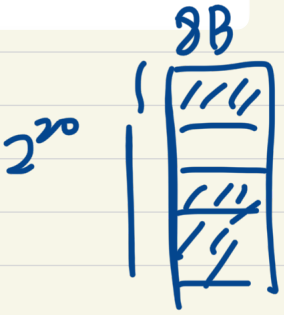
d) Given the parameters in the table above, calculate the total page table size for a system running 5 applications that utilize half of the memory available.

e) Given the parameters in the table above, calculate the total page table size for a system running 5 applications that utilize half of the memory available, given a two-level page table approach with 256 entries. Assume each entry of the main page table is 6 bytes. Calculate the minimum and maximum amount of memory required.

f) A cache designer wants to increase the size of a 4 KiB virtually indexed, physically tagged cache. Given the page size listed in the table above, is it possible to make a 16 KiB direct-mapped cache, assuming 2 words per block? How would the designer increase the data size of the cache?

解: (d)

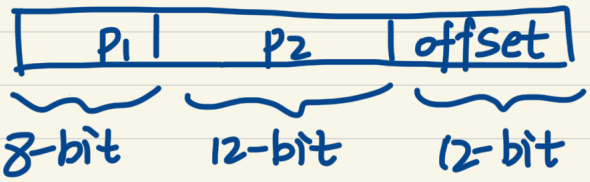


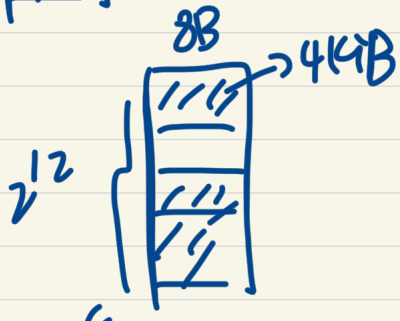


PTE Number $= 2^{20}$

Total Page Table Size $= 5 \times 8 \times 2^{20} = 40\text{MiB}$

(e)





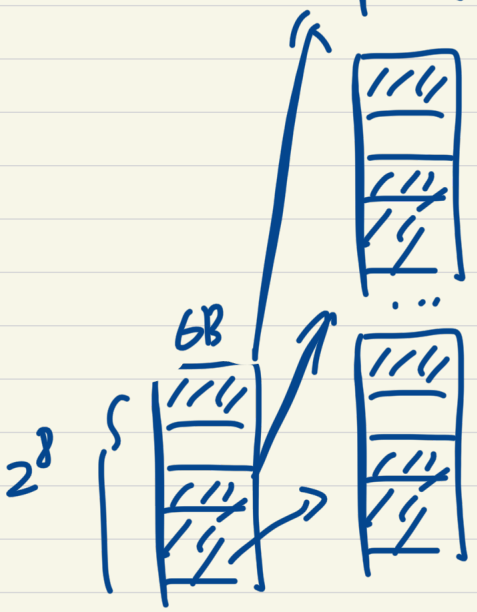
L1 Page Table Size: $6 \times 2^8 = 1.5\text{KiB}$

L2 Page Table Size: $8 \times 2^{12} = 32\text{KiB}$

L2页表数量$_{min} = 2^{31-12-12} = 2^7$

放 Memory $min = 5 \times (6 \times 2^8 + 2^7 \times 8 \times 2^{12})$

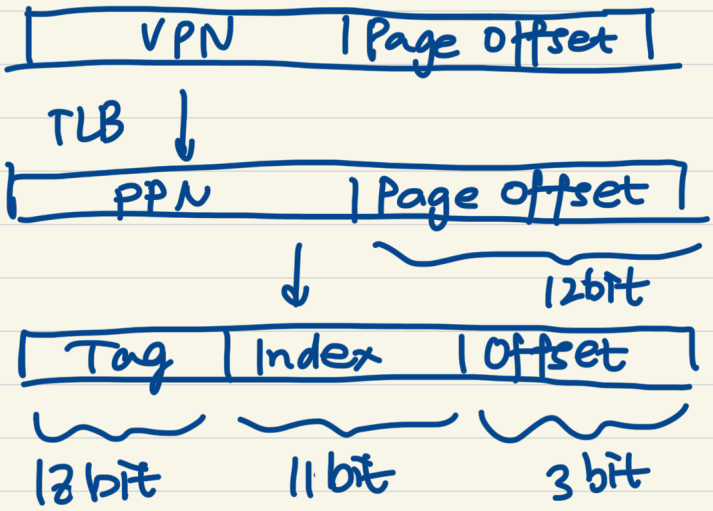
$= 20\text{MiB} + 7.5\text{KiB}$

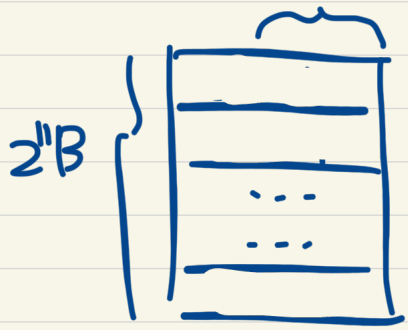


L2页表数量$_{max} = 2^{32-12-12} = 2^8$

放 Memory $max = 5 \times (6 \times 2^8 + 2^8 \times 8 \times 2^{12}) = 40\text{MiB} + 7.5\text{KiB}$

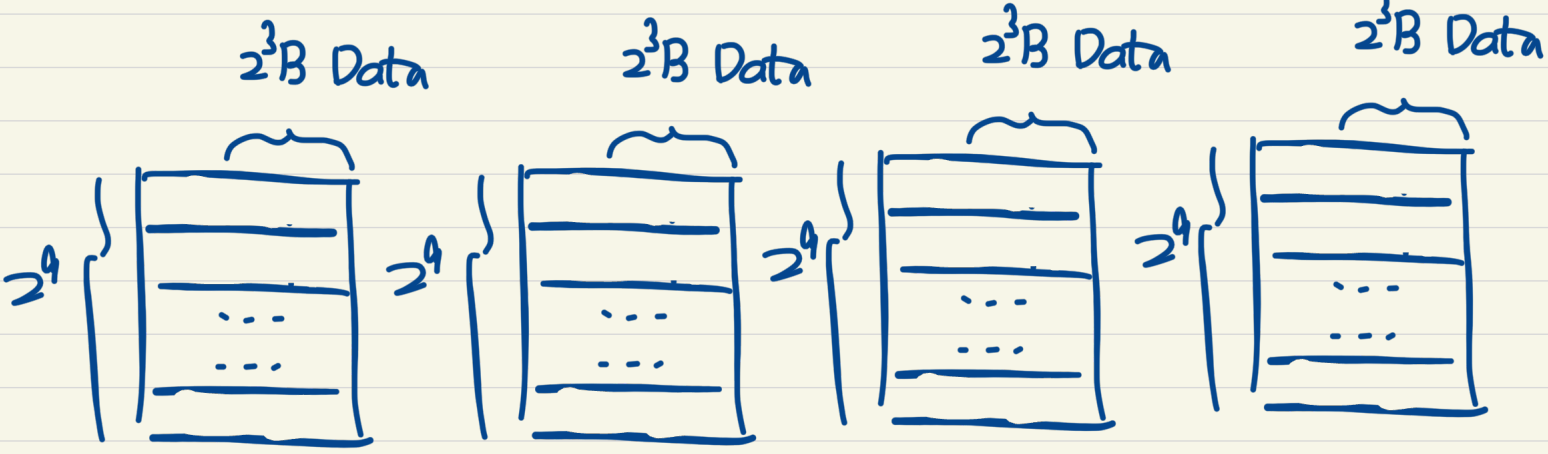
(f) 16KiB (2^{14}B) direct-mapped cache:



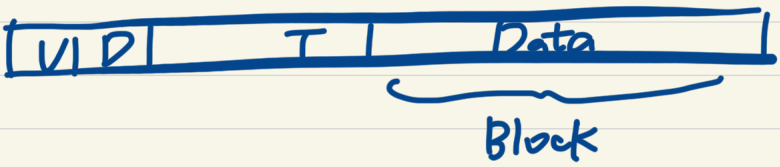


11bit+3bit=12bit 不可行

要将Index压缩为9-bit，即设计为4路组相联Cache



Cache Line



Size: $2^{\text{Offset(bit)}}$ Byte

Problem 2

Consider a system with 40-bit virtual addresses, 36-bit physical addresses, and 64 KiB (2^{16} bytes) pages. The system uses a page map to translate virtual addresses to physical addresses; each page map entry includes dirty (D) and resident (R) bits.

a) Assuming a flat page map, what is the size of each page map entry, and how many entries does the page map have?

Size of page map entry in bits: 22

Number of entries in the page map: 2^{24}

b) If changed the system to use 16 KiB (2^{14} bytes) pages instead of 64 KiB pages, how would the number of entries in the page map change? Please give the ratio of the new size to the old size.

(# entries with 16 KiB pages) / (# entries with 64 KiB pages): 4

c) Assume we still use 64 KiB pages. The contents of the page map and TLB are shown to the right. The page map uses an LRU replacement policy and the LRU page (shown below) will be chosen for replacement. For each of these four accesses, compute its corresponding physical address and indicate whether the access causes a TLB miss and/or a page fault. Assume each access starts with the TLB and Page Map state shown to the right.

TLB

VPN (tag)	V	D	PPN
0x0	1	0	0xBE7A
0x3	0	0	0x7
0x5	1	1	0xFF
0x2	1	0	0x900

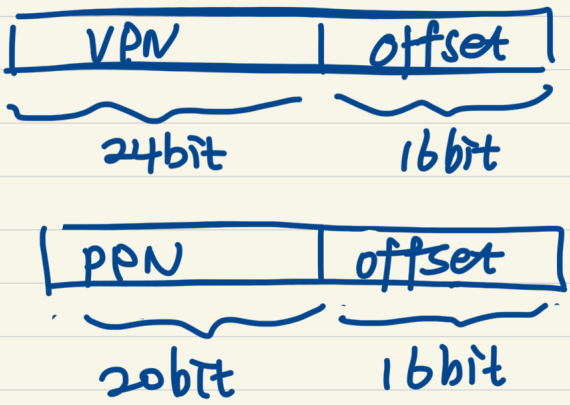
Fill in table below

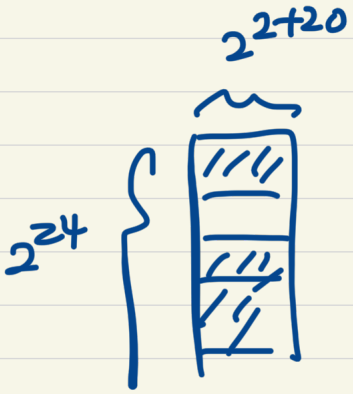
	Virtual Addr	PPN (in hex)	PhysAddr (in Hex)	TLB Miss? (Y/N)	Page Fault? (Y/N)
1.	0x06004	0xBE7A	0xBE7A6004	N	N
2.	0x30286	0x8	0x80286	Y	N
3.	0x68030	0x70	0x708030	Y	N
4.	0x4BEEF	0x8	0x8BEEF	Y	Y

Page Map

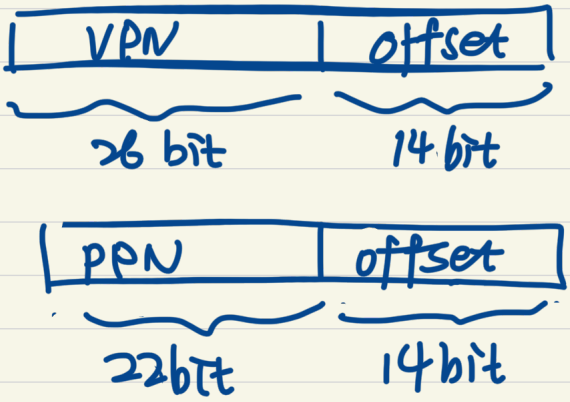
VPN	R	D	PPN	
0	1	0	0xBE7A	
1	0	0	---	
2	1	0	0x900	
3	1	0	0x8	⇦ LRU PAGE
4	0	0	---	
5	1	1	0xFF	
6	1	0	0x70	

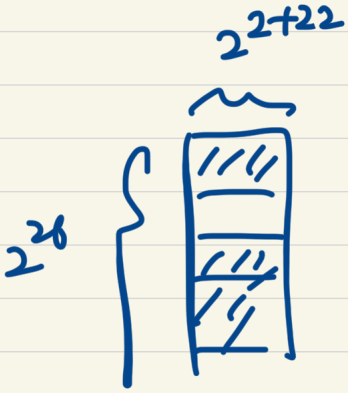
解: (a)


VPN
offset
24bit
16bit
PPN
offset
20bit
16bit

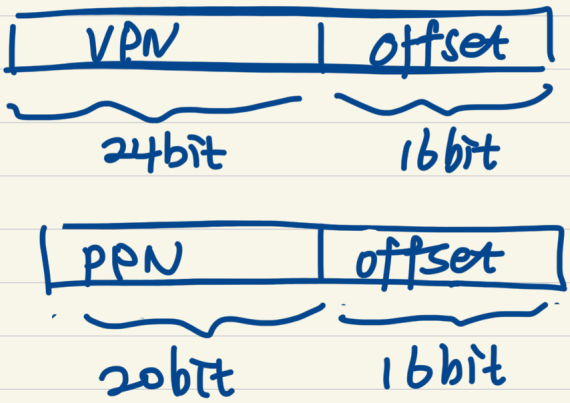

2^{2+20}
2^{24}

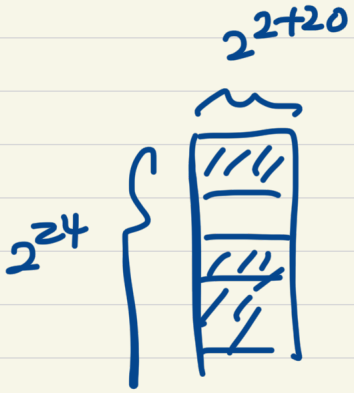
(b)


VPN
offset
26 bit
14 bit
PPN
offset
22bit
14bit


2^{2+22}
2^{26}

(c)


VPN
offset
24bit
16bit
PPN
offset
20bit
16bit


2^{2+20}
2^{24}

Problem 3

For a direct-mapped cache design with a 32bit address, the following bits of the address are used to access the cache.

Tag	Index	Offset
31-13	12-6	5-0
19 bit	7bit	6bit

a) What is the cache line size(in words)?

b) How many entries does the cache have?

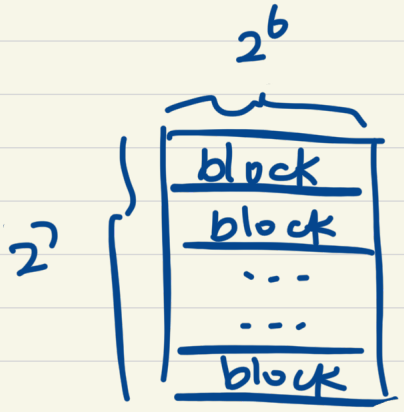
c) What is the ratio between total bits required for such a cache implementation over the data storage bits? (Note the structure of the direct-mapped cache! Don't forget the valid bit.)

解: (a) 2^6 Byte, 16words

Cache line = Cache Block

(b) $2^7 = 128$ entries

(c) $\text{Ratio} = \frac{2^7 \times (1+19+2^6 \times 2^3)}{2^7 \times 2^6 \times 2^3} = \frac{133}{128}$





Problem 4

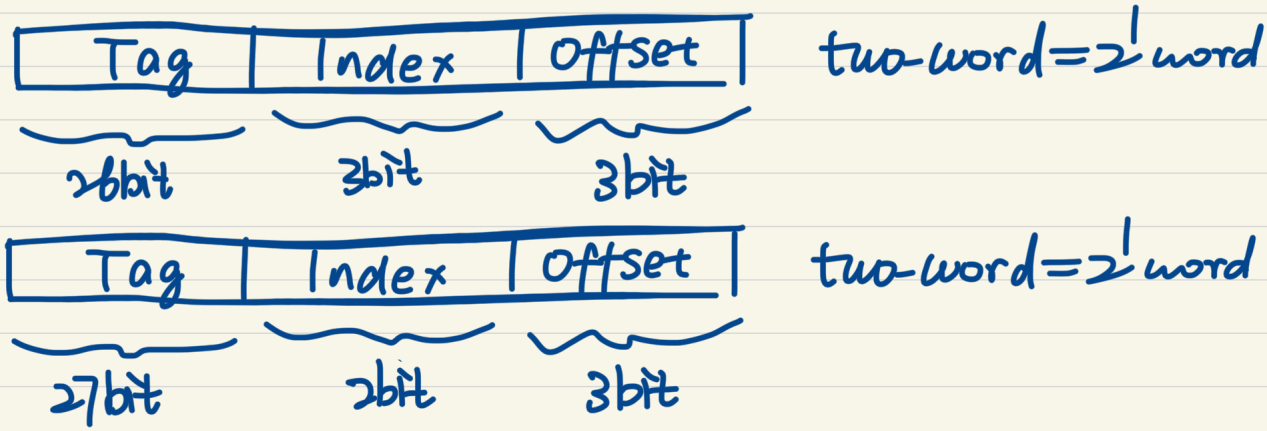
Cache is important to providing a high-performance memory hierarchy to processors. Below is a list of 32-bit memory address references, **given as word address**.
3, 180, 43, 2, 181, 88, 164, 180, 89.

(a) For each of these references, identify the binary address, the tag, and the index given a direct-mapped cache with 8 two-word blocks. Also list if each reference is a hit or a miss, assuming the cache is initially empty.

Word address	Binary address	Tag	Index	Hit/Miss
3	0000 0011	0	1	MISS
180	1011 0100	11	2	MISS
43 32+8+2+1	0010 1011	2	5	MISS
2	0000 0010	0	1	HIT
181 128+32+16+4+1	1011 0101	11	2	HIT
88 64+16+8	0101 1000	5	4	MISS
164 128+32+4	1010 0100	10	2	MISS
180 128+32+16+4	1011 0100	11	2	MISS
89 64+16+8+1	0101 1001	5	4	HIT

(b) For each of these references, identify the binary address, the tag, and the index given a direct-mapped cache with 4 four-word blocks. Also list if each reference is a hit or a miss, assuming the cache is initially empty.

Word address	Binary address	Tag	Index	Hit/Miss
3	0000 0011	0	1	MISS
180	1011 0100	22	2	MISS
43	0010 1011	5	1	MISS
2	0000 0010	0	1	HIT
181	1011 0101	22	2	HIT
88	0101 1000	11	0	MISS
164	1010 0100	10	2	MISS
180	1011 0100	22	2	MISS
89	0101 1001	11	0	HIT



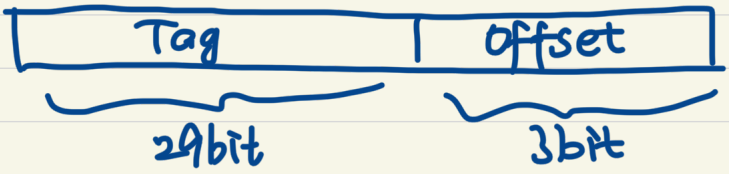
Problem 5

This exercise examines the impact of cache designs. For this exercise, the table of **word address** streams is shown in below.

1, 134, 212, 1, 135, 213, 162, 161, 2, 44, 41, 221

What is the miss rate for a fully associative cache with two-word blocks and a total size of 8 words, using LRU replacement?

解



Tag = word address / 2

Word Address	1	134	212	1	135	213	162	161	2	44	41	221
Tag	0	67	106	0	67	106	81	80	1	22	20	110
H/M	M	M	M	H	H	H	M	M	M	M	M	M

miss rate = $\frac{9}{12} = \frac{3}{4}$

Problem 6

You are trying to reverse-engineer the characteristics of a cache in a system, so that you can design a more efficient, machine-specific implementation of an algorithm you are working on. To do so, you have come up with three patterns that access various bytes in the system in an attempt to determine the following four cache characteristics:

- Cache block size (8, 16, 32, 64, or 128 B)
- Cache associativity (1-, 2-, 4-, or 8-way)
- Cache size (4 or 8 KB)
- Cache replacement policy (LRU or FIFO)

However, the only statistic that you can collect on this system is cache hit rate after performing the
access pattern. Here is what you observe (the address is in bytes):

Sequence	Addresses Accessed (Oldest → Youngest)								Hit Rate
1.	0	4	8	16	64	128			1/2
2.	31	8192	63	16384	4096	8192	64	16384	5/8
3.	32768	0	129	1024	3072	8192			1/3

Assume that the cache is initially empty at the beginning of the first sequence, but not at the
beginning of the second and third sequences. The sequences are executed back-to-back, i.e., no other
accesses take place between the three sequences. Thus, at the beginning of the second (third)
sequence, the contents are the same as at the end of the first (second) sequence.

Based on what you observe, what are the following characteristics of the cache?

Explain the reason.

(a) Cache block size (8, 16, 32, 64, or 128 B)?
(b) Cache associativity (1-, 2-, 4-, or 8-way)?
(c) Cache size (4 or 8 KB)?
(d) Cache replacement policy (LRU or FIFO)?

解: 由条件 Offset: 3bit~7bit Set bit: 0bit~3bit

Index + Offset + set number = 12bit~13bit

Address	Binary Address					Hit Number
0	0000	0000	0000	0000	M	
4	0000	0000	0000	0100	H	3Hit 3Miss
8	0000	0000	0000	1000	H	
16	0000	0000	0001	0000	H	
64	0000	0000	0100	0000	M	
128	0000	0000	1000	0000	M	
31	0000	0000	0001	1111	H	
8192	0010	0000	0000	0000	M	
63	0000	0000	0011	1111	M	
16384	0100	0000	0000	0000	H	5Hit 3Miss
4096	0001	0000	0000	0000	M	
8192	0010	0000	0000	0000	H	
64	0000	0000	0100	0000	H	
16384	0100	0000	0000	0000	H	
32768	1000	0000	0000	0000	M	
0	0000	0000	0000	0000	M	
129	0000	0000	1000	0001	H	2Hit 4Miss
1024	0000	0100	0000	0000	M	
3072	0000	1100	0000	0000	M	
8192	0010	0000	0000	0000	H	

① Offset=7bit时, Seq.1 Hit=4, 舍

② offset≤4bit时, Seq.1 Miss≥4, 舍

③ Offset=5bit时, Seq.2 2,3,4,5为Miss, 舍

故 Offset=6bit, Index+Set bit=6bit~7bit, Index: 3bit~7bit

1. 若 Index=7bit

Set bit=0 bit, Seq.2 Miss≥4, 舍

2. 若Index=6bit

Set bit≤1 bit, Seq.2 Miss≥4, 舍

3. 若Index=5bit

1) 若Set bit=1bit, Seq.2 Miss≥4, 舍

2) 若Set bit=2bit, 若FIFO, Seq.3 1,2,4,5,6为Miss, 舍

若LRU, 成立.

此时, Seq.1 2,3,4 Hit, Seq.2 2,3,5 Miss, Seq.3 3,6 Hit

4. 若Index=4bit

1) 若Set bit=2bit, Seq.3 仅3 Hit, 舍

2) 若Set bit=3bit, Seq.3 2,3,6为Hit, 舍

5. 若Index=3bit

Set bit≥3bit, Seq.3 2,3,6为Hit, 舍

综上, Offset=6bit, Index=5bit, Set bit=2bit

故a) Cache block size = 2^6 = 64B

b) 4-way associativity

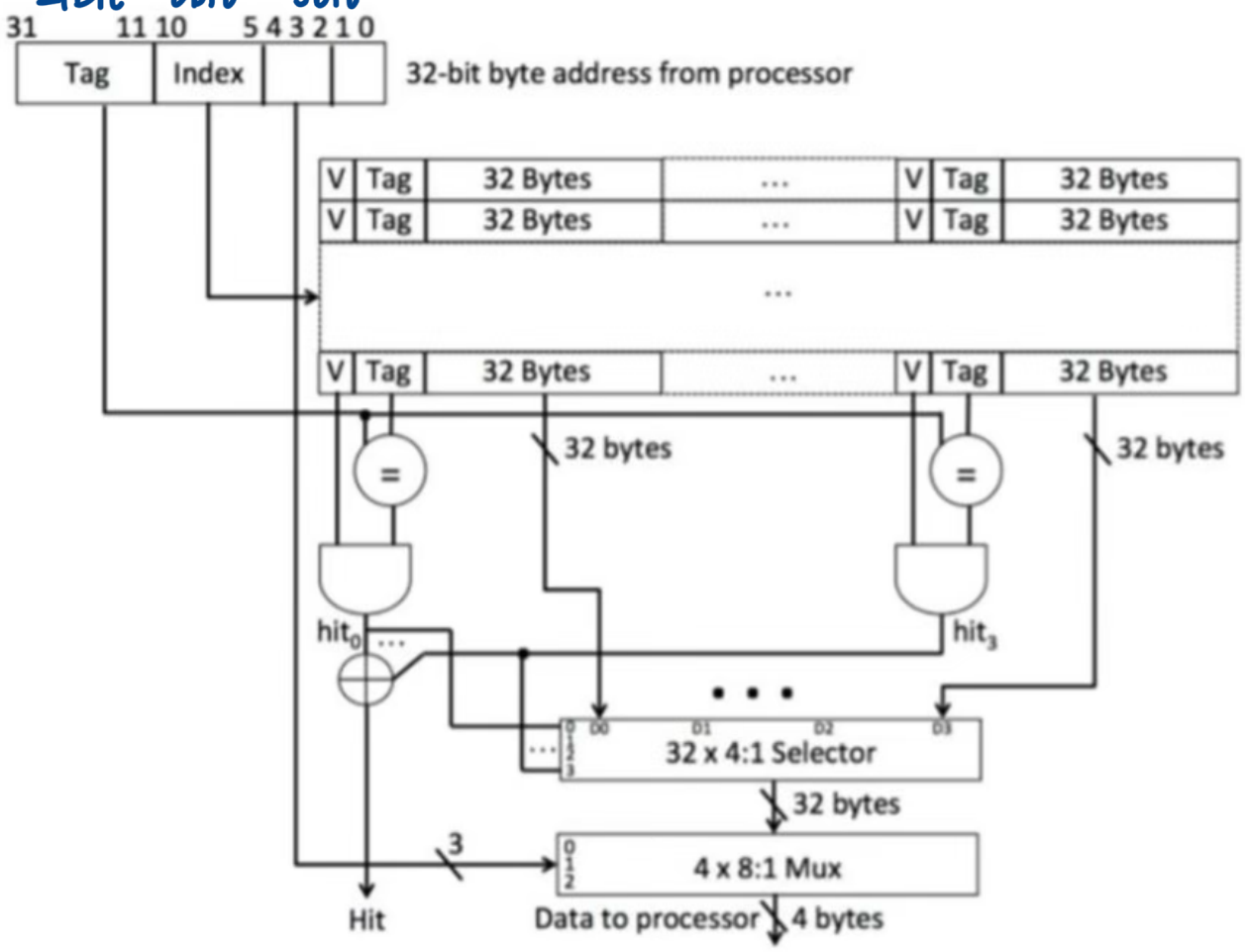
c) 8KB

d) LRU

Problem 7

You are given the sketch of a cache design below:

21bit 6bit 5bit



Note: a 4:1 Selector has four control inputs labeled 0, 1, 2, 3 and four data inputs D0, D1, D2, D3. It connects Di to the output when the hit control signal is true. Assume that at most one control input is true at any time. The selector function is undefined when none of the control inputs are true.

Answer the following questions about the cache above.

a) What is the block size of the cache in bytes? 32 Byte
b) What is the number of blocks in this cache? $2^6 \times 4 = 256$
c) What is the total data capacity of the cache in bytes? $256 \times 32 = 8192$ Byte
d) What is the associativity of the cache? 4-way set associative
e) Is this a write-through or write-back cache? write-through
f) What is the total number of valid bits in the cache? $4 \times 2^6 = 256$ bits
g) What is the total number of tag bits in the cache? $4 \times 2^6 \times 21 = 5376$ bits

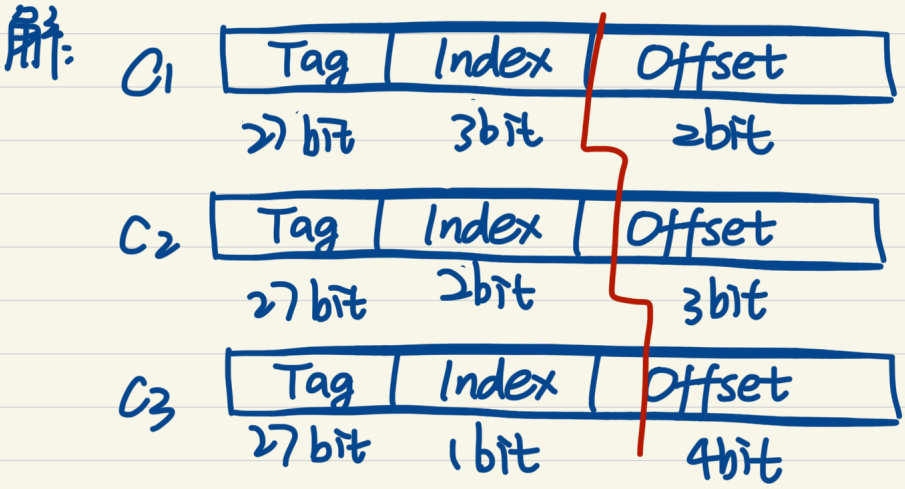
Problem 8

Caches are important to providing a high performance memory hierarchy to process. Below is a list of 32bit memory address references, **given as word address.**
7, 215, 175, 215, 7, 84, 65, 174, 64, 105, 85, 215

a) You're asked to optimize a cache design for the given references. There are three direct-mapped cache designs possible, all with total of 8 words of data:
C1 has 1-word blocks, C2 has 2-word blocks, and C3 has 4-word blocks.

In term of miss rate, which cache design is the best? If the miss stall time is 25 cycles, and C1 has an access time of 2 cycles, C2 takes 3 cycles, and C3 takes 5 cycles, which is the best cache design?

解:



%8 → /2 → /2%4 → /2

XXX XXX XXX

Word Address	7	215	175	215	7	84	65	174	64	105	85	215
Tag	0	26	21	26	0	10	8	21	8	13	10	26
Index C1	7	7	7	7	7	4	1	6	0	1	5	7
Hit/MISS	M	M	M	M	M	M	M	M	M	M	M	M
Index C2	3	3	3	3	3	2	0	3	0	0	2	3
Hit/MISS	M	M	M	M	M	M	M	M	H	M	H	M
Index C3	1	1	1	1	1	1	0	1	0	0	1	1
Hit/MISS	M	M	M	M	M	M	M	M	H	M	M	M

	C1	C2	C3
miss rate	1	5/6	11/12
Total cycle	324	286	335

Total Cycle = 12 x [miss rate x Stall time + access time]

无论从 miss rate 还是 Total cycle 角度, C2 is the best cache design